Interpretable Machine Learning

Functional ANOVA

Learning goals

- Understand one classical kind of functional decomposition: functional ANOVA (fANOVA).

...

- Algorithm for calculating the components in a fANOVA

??? HISTORY OF FANOVA ?????

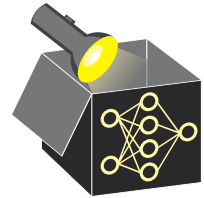
@ Sobol-Hoeffding Decomposition: erstes Paper von (Sobol? Hoeffding?) in 1950, nur im Unit Cube, allererste HDMR

FRAME TITLE

Assumption: Independent features

Today, we look at one method to obtain a functional decomposition; Funcational ANOVA (fANOVA).

IDEA

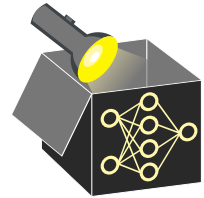


Consider example from before:

$$\hat{f}(x_1, x_2) = 4 - 2x_1 + 0.3e^{x_2} + |x_1|x_2 \tag{1}$$

Goal: Interaction term $g_{1,2}$ should only contain the pure interaction, in this case only $|x_1|x_2$, all the rest are actually main effects.
General idea: Try to move as much as possible into lower-order effects, and only keep the rest in the higher-order term.
$\Rightarrow$ First compute the lower-order terms, then the higher order terms.

IDEA



$$\hat{f}(x_1, x_2) = 4 - 2x_1 + 0.3e^{x_2} + |x_1|x_2 \tag{2}$$

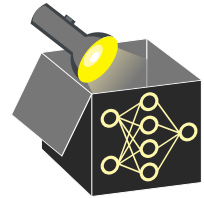
Question: How to compute lower-order terms?
Starting with the main effects: *Link to PDP??*
$\Rightarrow$ fANOVA uses the same idea as PDPs, but also for higher-order terms: the resp. feature is fixed and all other features are simply averaged out.
Total formula for calculating the components g_S in the fANOVA algorithm:

$$g_S(\mathbf{x}_S) = (\text{average out all features not contained in } S) - (\text{All lower-order components})$$

idea: PD-function = sum of all g_S up to this order

FORMAL DEFINITION AND COMPUTATION

Hooker (2004)



Components are recursively defined using marginals (here $-S = \{1, \ldots, p\} \setminus S$ denotes the set of all indizes not contained in S):

$$g_S(\mathbf{x}_S) = \mathbb{E}_{\mathbf{X}_{-S}} \left[\hat{f}(\mathbf{x}_S, \mathbf{X}_{-S}) \,\middle|\, \mathbf{x}_S \right] - \sum_{V \subset S} g_V(x_V)$$

- Expectation integrates $\hat{f}(\mathbf{x})$ over all input features except $\mathbf{x}_S$
- Subtract all components g_V with $V \subset S$ to remove all lower-order effects and center the effect

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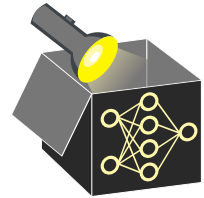

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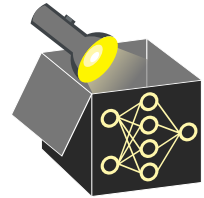
- Expectation integrates $\hat{f}(\mathbf{x})$ over all input features except $\mathbf{x}_S$
- Subtract all components g_V with $V \subset S$ to remove all lower-order effects and center the effect
- Recursive computation:

$$\begin{aligned}
g_\emptyset &= \mathbb{E}_X\left[\hat{f}(\mathbf{x})\right] \\
g_j(x_j) &= \mathbb{E}_{X_{-j}}\left[\hat{f}(\mathbf{x}) \mid x_j\right] - g_\emptyset, \ \forall j \in \{1, \ldots, p\} \\
g_{j,k}(x_j, x_k) &= \mathbb{E}_{X_{-\{j,k\}}}\left[\hat{f}(\mathbf{x}) \mid x_j, x_k\right] - g_k(x_k) - g_j(x_j) - g_\emptyset, \ \forall j < k \\
&\vdots \\
g_{1,\ldots,p}(\mathbf{x}) &= \hat{f}(\mathbf{x}) - g_{1,\ldots,p-1}(x_1, \ldots x_{p-1}) - \cdots - g_{1,2}(x_1, x_2) - g_p(x_p) - \cdots - g_2(x_2) - g_1(x_1) - g_\emptyset \\
&= \hat{f}(\mathbf{x}) - \textstyle\sum_{S \subsetneq \{1,\ldots,p\}} g_S(\mathbf{x}_S)
\end{aligned}$$

EXAMPLE

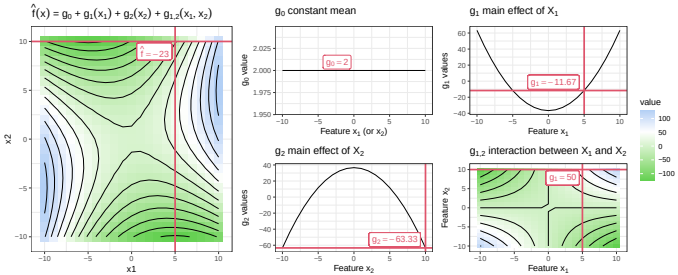
[calculate a full complete example]
Analytic examples from Sobol (1993), Sobol (2001) or Hooker(2004))
[several examples needed, can also include those from beginning]
[problem with examples from beginning: If using fANOVA, we do not get the original components out of it again, right?]
-> exactly that is the case, explain the reason here: some averaged part of interaction term is moved into lower order, we need this in fANOVA, because we need a data-driven / formula-agnostic way of computing the components
-> use the in-class calculation task here to illustrate this

FUNCTIONAL DECOMPOSITION – EXAMPLE



Example: $\hat{f}(\mathbf{x}) = 2 + x_1^2 - x_2^2 + x_1 \cdot x_2$ (e.g., if $x_1 = 5$ and $x_2 = 10 \Rightarrow \hat{f}(\mathbf{x}) = -23$)

- Computation of components using feature values $x_1 = x_2 = (-10, -9, \ldots, 10)^\top$ gives:



For $x_1 = 5$ and $x_2 = 10$:

- $g_\emptyset = 2$
- $g_1(x_1) = -9.67$
- $g_2(x_2) = -65.33$
- $g_{1,2}(x_1, x_2) = 50$

$\Rightarrow \hat{f}(\mathbf{x}) = -23$

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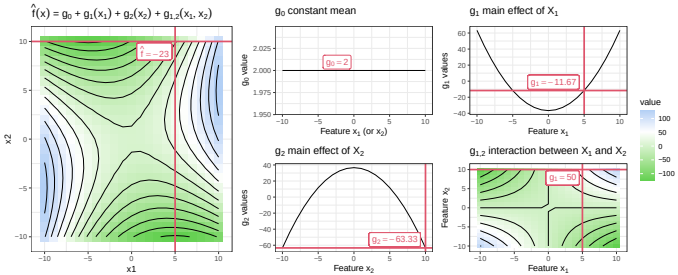

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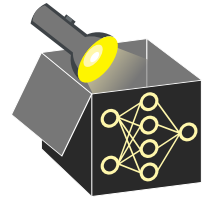
- Vanishing condition means:
  - g_1 and g_2 are mean-centered w.r.t. marginal distribution of x_1 and x_2
  - Integral of $g_{1,2}$ over marginal distribution x_1 (or x_2) is always 0.

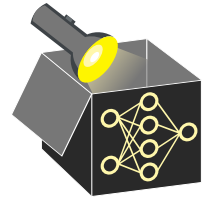
HOW TO COMPUTE FANOVA IN PRACTICE $\rightarrow$ ESTIMATING EXPECTATIONS



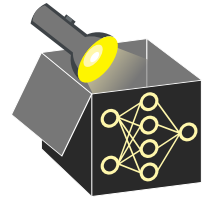
Estimating expectations over uniform distribution using e.g. Monte-Carlo sampling

[explain Monte-Carlo integration]

EXAMPLE: MORE REALISTIC CALCULATION ?



VARIANCE DECOMPOSITION - WHY IS THIS METHOD EVEN CALLED FUNCTIONAL ANOVA?



- Decomposition of $\hat{f}(\mathbf{x})$ allows to conduct functional analysis of variance (fANOVA)

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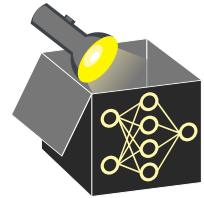

- Decomposition of $\hat{f}(\mathbf{x})$ allows to conduct functional analysis of variance (fANOVA)
- If features are independent, one can prove that with the components defined in the fANOVA, the variance can be additively decomposed without covariances:

$$
\begin{aligned}
Var\left[\hat{f}(\mathbf{x})\right] &= Var\left[g_{\emptyset} + g_1(x_1) + \ \ldots \ + g_{1,2}(x_1, x_2) + \ \ldots \ + g_{1,\ldots,p}(\mathbf{x})\right] \\
&= Var\left[g_{\emptyset}\right] + Var\left[g_1(x_1)\right] + \ \ldots \ + Var\left[g_{1,2}(x_1, x_2)\right] + \ \ldots \ + Var\left[g_{1,\ldots,p}(\mathbf{x})\right]
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In other words: The single components are uncorrelated.

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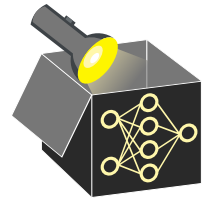

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- Dividing by the prediction variance, yields fraction of variance explained by each term:

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1 = \frac{Var\left[g_{\emptyset}\right]}{Var\left[\hat{f}(\mathbf{x})\right]} + \frac{Var\left[g_1(x_1)\right]}{Var\left[\hat{f}(\mathbf{x})\right]} + \ \ldots \ + \frac{Var\left[g_{1,2}(x_1, x_2)\right]}{Var\left[\hat{f}(\mathbf{x})\right]} + \ \ldots \ + \frac{Var\left[g_{1,\ldots,p}(\mathbf{x})\right]}{Var\left[\hat{f}(\mathbf{x})\right]}
$$

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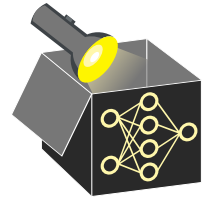

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- Fraction of variance explained by a component $g_V(\mathbf{x}_V)$ is the Sobol index:
  $S_V = \frac{Var[g_V(\mathbf{x}_V)]}{Var\left[\hat{f}(\mathbf{x})\right]}$
  $\rightsquigarrow$ Importance measure of component $g_V(\mathbf{x}_V)$
  $\rightsquigarrow$ Small S_V values $\Rightarrow$ Component g_V does not explain much of total variance of $\hat{f}$